HP VICTUS RYZEN 5 5600H

FORENSIC DATA ANALYSIS REPORT

Battery state, charging behavior, firmware/EC, ACPI, telemetry, cross-GPU evidence, and causal assessment

PRIMARY SUBJECT	HP Victus 15-fb0147AX / Ryzen 5 5600H / Radeon RX 6500M / Board 8A3F
INVESTIGATION WINDOW	September 2024 - August 2026
PRIMARY OBSERVATION	Intermittent, state-dependent battery/power-subsystem anomaly reproduced across Windows, Linux, and HP pre-boot diagnostics
ANALYTICAL STATUS	Strong basis for platform-level firmware/EC/ACPI investigation; exact failed component remains unresolved
PUBLICATION POSTURE	Evidence-led technical disclosure. No claim of universal defect, intentional conduct, or single proven root cause.

Source corpus includes the 18-page forensic report, 6-page condensed report, master battery/power evidence archive, ACPI/EC and BIOS reverse-engineering reports, raw Windows battery-report HTML files, and preserved diagnostic evidence.

PUBLIC TECHNICAL EDITION

Prepared 25 August 2026. This document is an analytical synthesis of supplied evidence. External community cases are identified as source-derived unless independently verified in this edition.

1. EXECUTIVE FINDINGS

The evidence supports a focused investigation into a shared Victus-platform battery/power-management mechanism on the Ryzen 5 5600H platform. The strongest evidence does not depend on a single Windows percentage reading: it combines raw battery-report history, controlled WMI state transitions, independent Linux UPower/sysfs observations, HP UEFI diagnostics, and a decompiled ACPI/EC battery path.

Cross-GPU recurrence: The supplied cross-GPU matrix covers five discrete GPU configurations. Rapid discharge, charging halt, and hybrid drain have non-zero signals in all five; telemetry corruption, ACPI/BIOS error, and power lockout have non-zero signals in four of five. This is pattern evidence, not prevalence data.

State transition anomaly: Controlled WMI data show remaining-energy reporting moving from 0.300 Wh to about 50.184 Wh after an EC/power reset while charging remained inactive; full-charge capacity changed only 34 mWh.

Independent OS corroboration: Fedora reproduced near-empty and near-full states on the same platform, weakening a Windows-only explanation.

OEM diagnostic contradiction: HP diagnostics moved from Normal/OK at high SOC to Primary Battery: Replace at low SOC, with a recorded failure identifier. The same-day diagnostic history includes repeated PASS results followed by FAIL results.

Firmware architecture: DSDT analysis establishes a BMS/fuel-gauge -> SMBus -> EC -> ACPI -> OS data path and identifies battery-related EC-backed fields consumed by _BST/_BIF/_BIX methods.

Bottom-line assessment

Strongly supported: a genuine, intermittent/state-dependent battery/power-subsystem anomaly exists on the investigated unit, and the anomaly is not adequately explained by a Windows-only software problem. Supported hypothesis: a shared firmware/EC/ACPI/power-management mechanism merits investigation across 5600H Victus configurations. Not established: a universal Victus defect, a unique EC root cause, a specific BIOS revision as the cause, or intentional OEM conduct.

2. DATA CORPUS, LINEAGE, AND EVIDENCE HIERARCHY

The analysis treats original artifacts as higher-value than derived summaries. The underlying record itself notes that raw screenshots and raw battery-report HTML should control where a transcription or OCR conflict exists.

ID	Source	Role	Status
P1	battery-report.html	Raw Windows battery telemetry; 75 numeric FCC observations in this file	Primary
P2	battery-report-2026-08-22.html	Fresh F.26 Windows battery report; 7 FCC observations; post-anomaly stabilization snapshot	Primary
P3	HP Victus Ultimate Consolidated Battery Evidence Report 2026-08-16	Device chronology, HP diagnostics, Fedora, battery transitions, evidence index	Secondary synthesis
P4	HP Victus Forensic Research Report (18 pages)	Cross-GPU synthesis, DSDT/EC architecture, causality limits, references	Secondary synthesis
P5	HP Victus Forensic Research Report-1 (6 pages)	Condensed cross-GPU matrix and public technical summary	Secondary synthesis
P6	HP Victus BIOS Reverse Engineering Status Report 2026-08-20	Firmware identity, F.26 / F.26 Rev.A reconciliation, reverse-engineering gate	Technical status
P7	ACPI/EC investigation reports	ACPI/EC architecture mapping and observed firmware issues	Technical analysis
P8	HP Master Diagnostic / Final Technical Evidence reports	HP test history, raw diagnostic interpretation, chronology	Secondary synthesis
P9	External HP Community cases	Independent corroborating cases; selected cases were web-verified for this edition	External

Snapshot mismatch is data, not noise: the raw battery reports differ in cycle count and reported design/FCC values because they are different captures. One raw file shows 84 cycles and 47.562 Wh in its current battery header; the later F.26 report shows 85 cycles and 49.480 Wh. The report does not merge these into one artificial value. Instead, the difference is treated as part of the telemetry history.

External-source caveat: community cases are useful for corroboration and hypothesis generation, but they are not controlled experiments. Their exact hardware, firmware, and battery conditions must be treated individually.

3. CROSS-GPU DATA ANALYSIS

The source evidence compares five discrete-GPU configurations on the Ryzen 5 5600H Victus platform. The matrix below is reproduced as a qualitative source-assigned severity coding: None=0, Rare=1, Common=2, Severe=3. It is not a prevalence estimate and cannot be used to calculate failure rates.

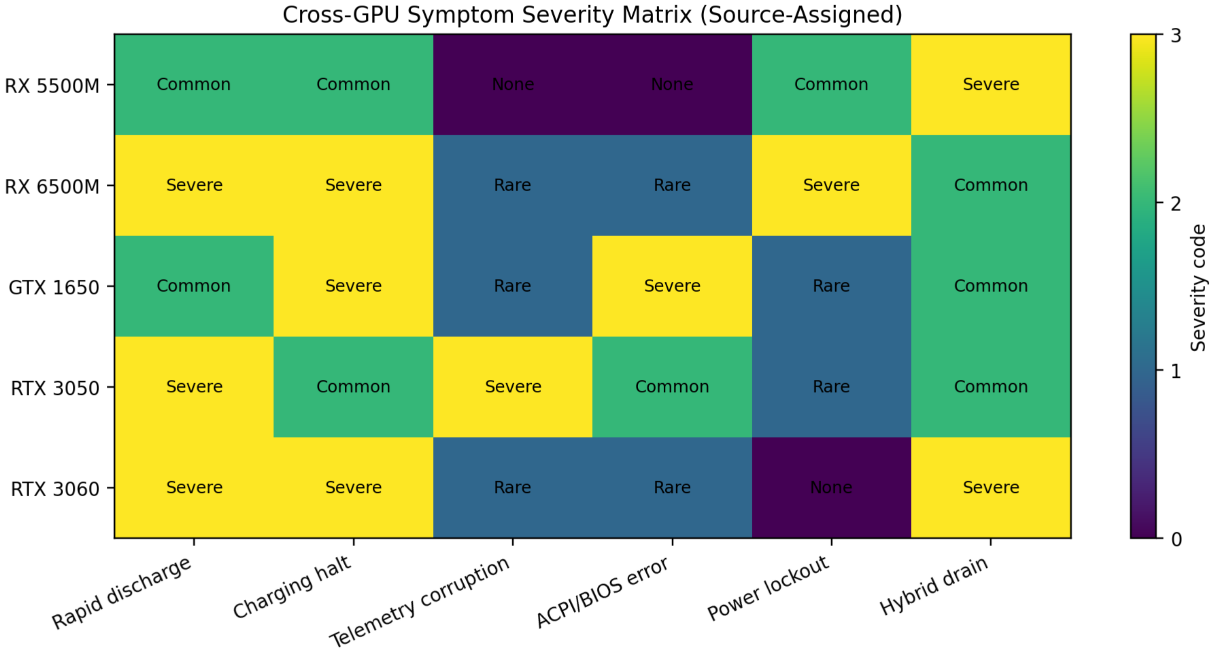


Figure 1. Source-assigned symptom severity across five GPU configurations. The meaningful analytic result is coverage: three core symptom classes are non-zero in all five configurations, while three additional classes are non-zero in four of five.

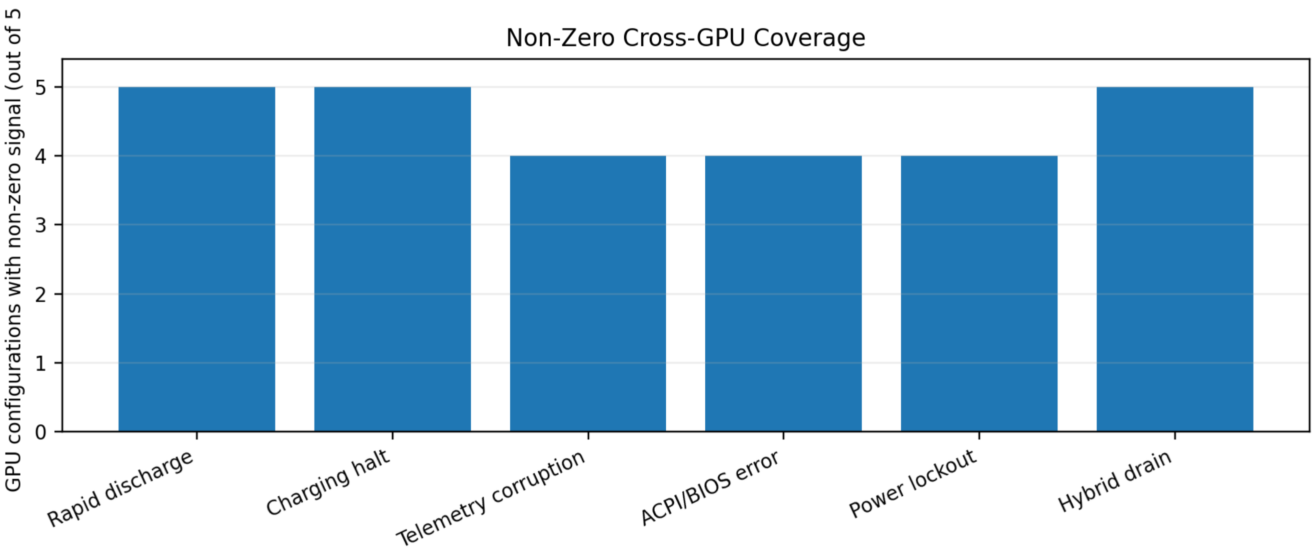


Figure 2. Non-zero coverage count. This demonstrates recurrence across GPU architectures, not incidence.

GPU	Rapid discharge	Charging halt	Telemetry	ACPI/BIOS	Power lockout	Hybrid drain
RX 5500M	Common	Common	None	None	Common	Severe
RX 6500M	Severe	Severe	Rare	Rare	Severe	Common
GTX 1650	Common	Severe	Rare	Severe	Rare	Common
RTX 3050	Severe	Common	Severe	Common	Rare	Common
RTX 3060	Severe	Severe	Rare	Rare	None	Severe

4. PRIMARY-UNIT CHRONOLOGY AND DATA INTEGRITY

Primary device: HP Victus 15-fb0147AX / Product 8F503PA#ACJ / Ryzen 5 5600H / Radeon RX 6500M. The battery manufacture date recorded by HP is 21 June 2024; the earliest available Windows battery history begins 18 September 2024. BIOS history records F.22 as an earlier baseline and F.26 on 20 March 2026, with EC version notes 35.48 -> 35.49.

Date	Observation	Analytical role
2024-06-21	HP battery manufacture date shown	Device chronology
2024-08	Owner-stated purchase period	Ownership chronology; not independently verified
2024-09-18	Earliest machine-level Windows battery history	Start of preserved telemetry record
2024-12-09	Reported FCC reaches 52.611 Wh in one source extraction	Non-monotonic reported capacity signal
2026-03-20	BIOS F.22 -> F.26; EC 35.48 -> 35.49	Firmware milestone; causation not established
2026-08-15 AM	Fedora reports ~50 Wh full/design, 85 cycles; coherent 63% -> 57% discharge	Cross-platform confirmation; real power draw present
2026-08-15 PM	Repeated implausible Windows transitions and Battery changed events	State/telemetry anomaly
2026-08-15 22:05	HP UEFI Battery Test: Primary Battery: Replace	OEM diagnostic evidence
2026-08-15 late	Windows reaches 0%/not charging while AC connected in one captured state	Charging-control/state anomaly
2026-08-16+	Service-level diagnosis and evidence consolidation	Escalation record
2026-08-20	Firmware RE project: exact March F.26 identity still unresolved	Important boundary before firmware attribution
2026-08-22	Fresh F.26 battery report shows 50.358 Wh FCC on 18-20 Aug and 50.346 Wh by 20-21 Aug	Post-reset stabilization snapshot

Version-control warning: the BIOS reverse-engineering report explicitly states that the March 20 F.26 baseline and a newer May 1 F.26 Rev.A package may not be byte-identical. The March baseline should therefore be preserved and fingerprinted before any firmware update or causal claim.

5. BATTERY CAPACITY / TELEMETRY TIME-SERIES ANALYSIS

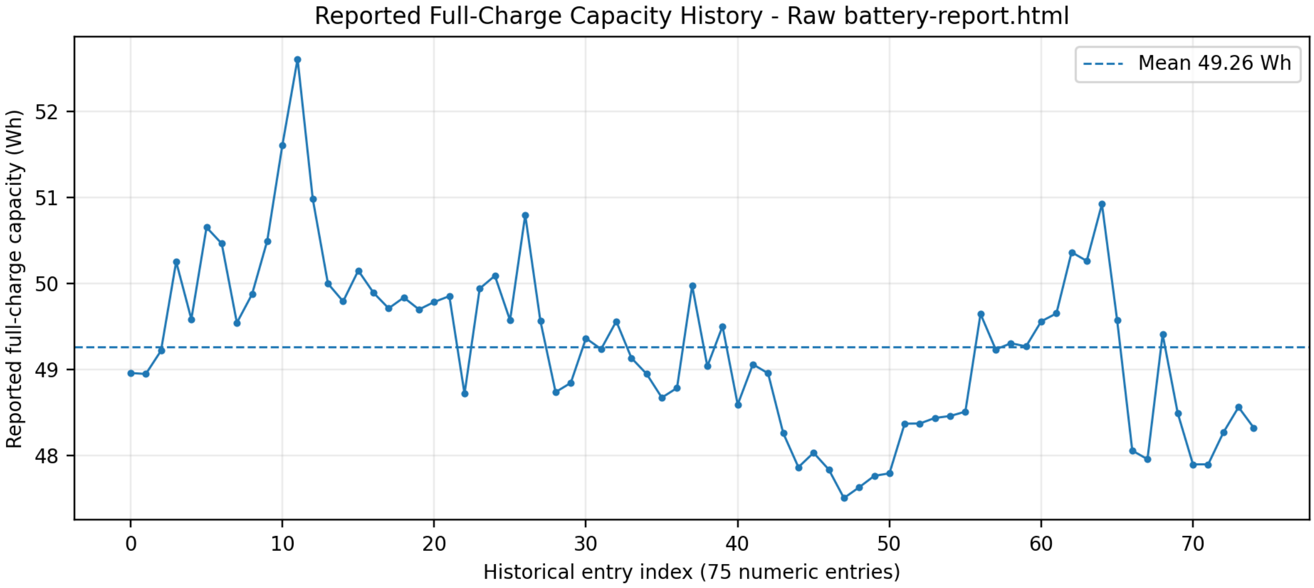


Figure 3. Raw battery-report.html FCC history. This capture contains 75 numeric capacity entries with a range of 47.505 to 52.611 Wh and a mean of 49.260 Wh. The report treats these as management/learned telemetry, not direct electrochemical measurements.

Snapshot	FCC (Wh)	Change vs prior	Interpretation
2024-09-18	48.959	Baseline	Earliest preserved record
2024-12-09	52.611	+3.652 Wh vs baseline	Reported capacity higher than initial value
2026-08-03	47.957	-4.654 Wh vs 2024 peak	Downward state change
2026-08-04	49.409	+1.452 Wh in one entry step	Large one-step increase in reported value
2026-08-10	48.323	-1.086 Wh from Aug 9	Continued variability
2026-08-15 fresh capture	49.777	Different report snapshot	Fresh F.26 report; not merged with prior capture
2026-08-21	50.346	+0.569 Wh vs Aug 15	Post-event stabilization near 50 Wh

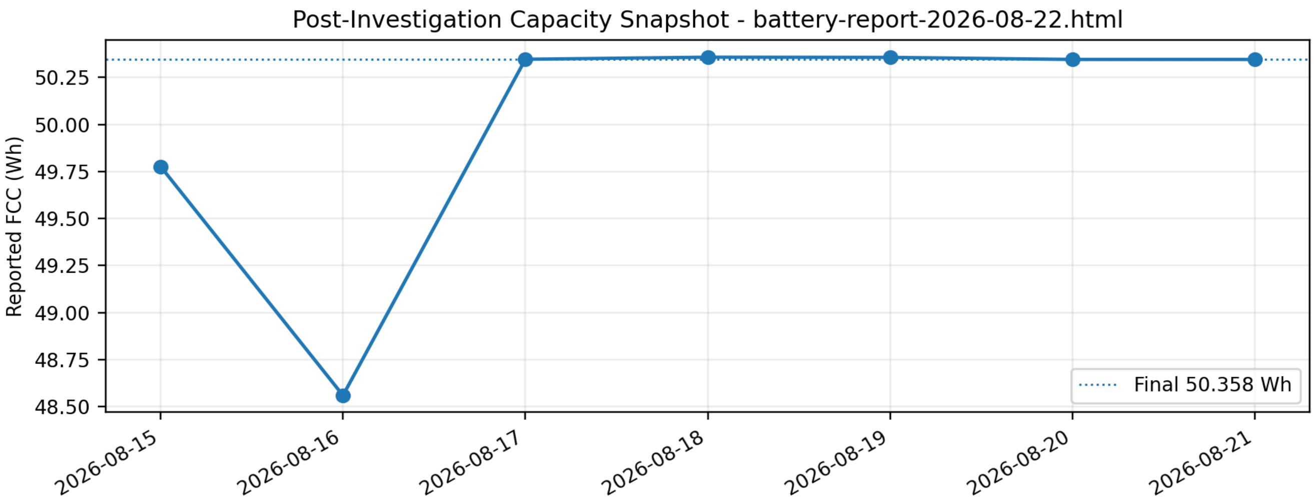


Figure 4. The later F.26 report shows a short post-event sequence: 49.777 Wh -> 48.559 Wh -> 50.347 Wh -> 50.358 Wh -> 50.357 Wh -> 50.346 Wh -> 50.346 Wh. This demonstrates that the reporting state can move materially over days and then settle near a stable value; it does not by itself identify the hardware component responsible.

6. ARITHMETIC AUDIT OF AUGUST 15 STATE TRANSITIONS

The supplied long report contains several source labels such as ~45.5 kW and ~42.0 kW. Those labels are arithmetically inconsistent with the stated energy deltas and durations. This edition recomputes the apparent rate directly as delta Wh / elapsed seconds, and preserves the raw delta/time values. The correction is important because it removes an avoidable credibility attack.

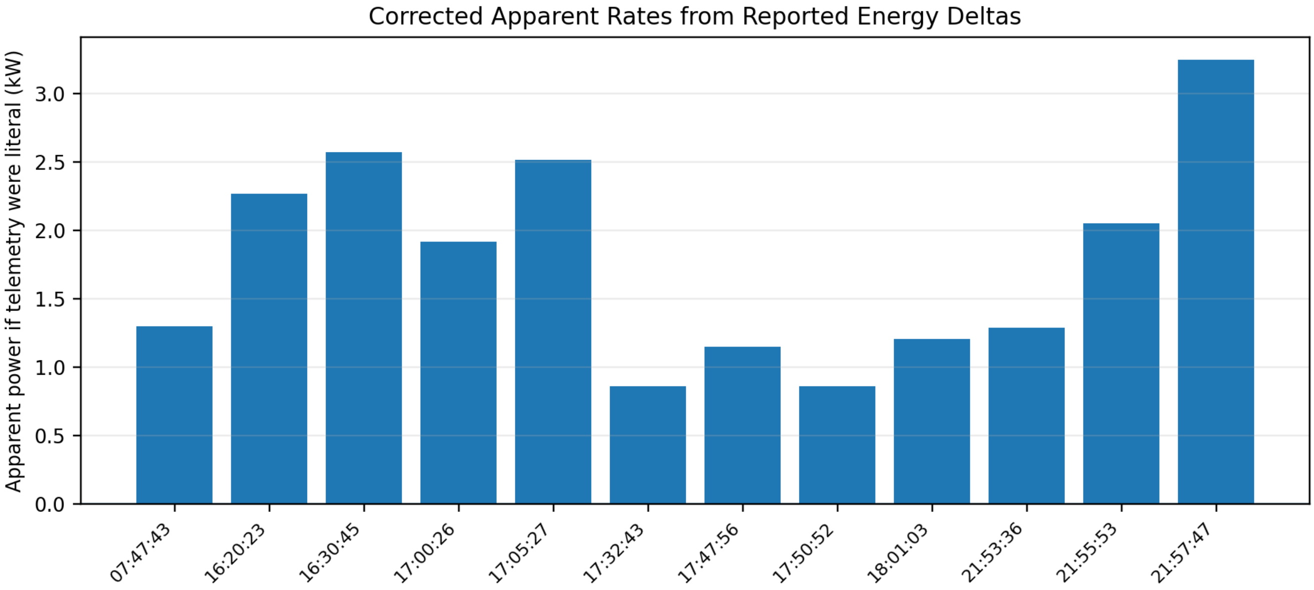


Figure 5. Corrected apparent rates from the source-reported energy deltas. These are diagnostic calculations only, not direct wattmeter measurements.

Time	Reported transition	Delta Wh	Duration	Recomputed apparent rate
07:47:43	52% -> 100%	24.151	67 s	1.30 kW
16:20:23	100% -> 34%	33.368	53 s	2.27 kW
16:30:45	100% -> 76%	12.139	17 s	2.57 kW
17:00:26	100% -> 84%	7.981	15 s	1.92 kW
17:05:27	100% -> 75%	11.169	16 s	2.51 kW
17:32:43	100% -> 90%	5.255	22 s	0.86 kW
17:47:56	100% -> 69%	15.627	49 s	1.15 kW
17:50:52	100% -> 49%	25.629	107 s	0.86 kW
18:01:03	100% -> 22%	38.855	116 s	1.21 kW
21:53:36	100% -> 46%	27.200	76 s	1.29 kW
21:55:53	100% -> 33%	33.587	59 s	2.05 kW
21:57:47	100% -> 12%	44.190	49 s	3.25 kW

Interpretation: corrected apparent rates span roughly 0.86-3.25 kW across the more complete event table. For a ~50 Wh laptop battery, the magnitude and timing are still physically implausible as literal chemical battery exchange. The defensible finding is a discontinuity, recalculation, or reclassification in reported battery state.

Control observation from the same Windows battery report: one normal-looking discharge session recorded about 34.453 Wh over 1 h 45 m 44 s, or approximately 19.6 W average consumption. This matters because the record contains both plausible physical power consumption and implausible state transitions; the anomaly is not simply "all telemetry is wrong".

7. CONTROLLED WMI STATE-TRANSITION AND LINUX CORROBORATION

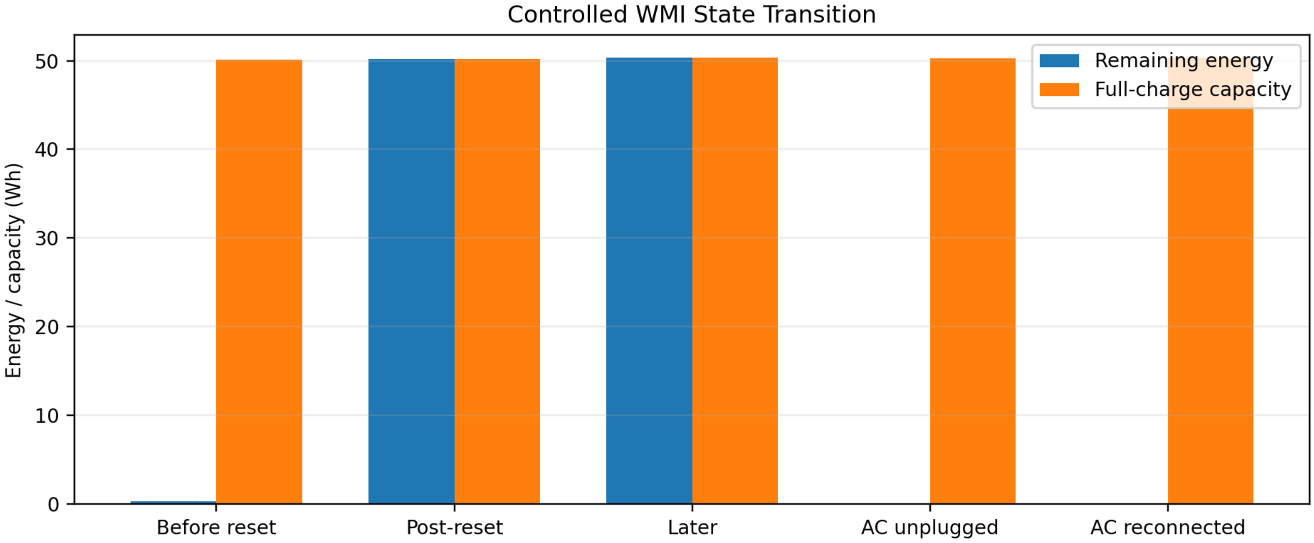


Figure 6. WMI remaining-energy and full-capacity values before/after the EC/power reset. Remaining energy changes by ~49.884 Wh while full-capacity changes by only 34 mWh from 50.150 to 50.184 Wh.

State	Remaining (Wh)	Full cap (Wh)	Voltage (V)	AC online	Charging
Low-SOC before reset	0.300	50.150	10.156	True	False
Post-reset	50.184	50.184	10.188	True	False
Later post-reset	50.358	50.358	9.934	True	False
AC unplugged	0.000	50.265	8.708	False	False
AC reconnected	0.000	50.358	9.095	True	False

Controlled-test inference: the evidence is most consistent with battery-state / fuel-gauge reinitialization or reclassification. It does not independently distinguish a battery fuel-gauge problem from EC-side handling, charging-controller behavior, or another component in the same management chain.

Linux corroboration: Fedora observations recorded a coherent session around 63% / 31.82 Wh to 57% / 28.655 Wh over about 10 minutes at roughly 40-47 W instantaneous power. The same investigation later documented near-empty states around 0.092 Wh and later near-full states around 50.173 Wh using Linux sysfs/UPower. The cross-platform value is not that Linux is "correct" in every state; it is that an independent OS path reproduced the same class of state discontinuity.

Low-workload context: supplied Windows Task Manager snapshots show representative investigation states with very low CPU/GPU utilization, which weakens a simple runaway-workload explanation for the largest instantaneous state discontinuities. This does not prove zero system power draw at every event.

8. HP MANUFACTURER DIAGNOSTIC STATE FLIP

The same HP diagnostic stack produced materially different outcomes depending on battery state. The source evidence records repeated PASSED results at higher SOC, followed by FAILED results and a Primary Battery: Replace result at low SOC.

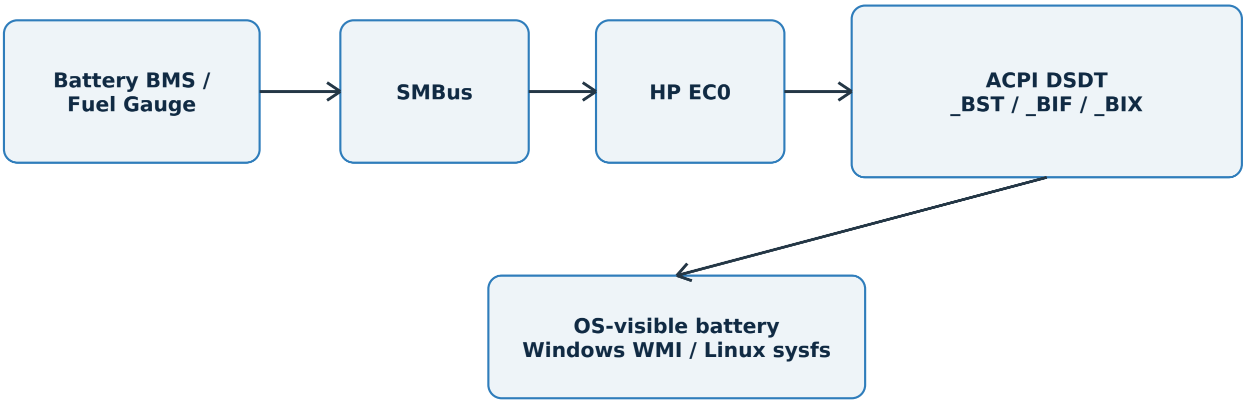
Time	HP diagnostic result	Context
21:50	PASSED	Battery Check
21:53	PASSED	Battery Check
21:56	PASSED	Battery Check
21:59	PASSED	Battery Check
22:17	FAILED	Battery Check
22:24	FAILED	Battery Check
22:27	FAILED	Battery Check
22:05	Primary Battery: Replace	Low-SOC UEFI check; Failure ID recorded

Key interpretation: the diagnostic inconsistency is strong evidence of a state-dependent battery-subsystem condition. It is not, by itself, proof that the EC is the failed component. The battery pack/BMS/fuel gauge and charging path remain live alternatives until service substitution or direct electrical capture is performed.

The failure identifier recorded in the original evidence is `6765U3-C80D73-MFPV7G-C0BQ23`. The source archive explicitly warns that OCR variants exist and the original screenshot should control.

9. FIRMWARE, EC, AND ACPI DATA-PATH ANALYSIS

ESTABLISHED DATA PATH



The mapped path establishes interfaces and data flow; it does not uniquely localize the failed component.

Field	Width	Role identified in source decompilation
BADC	16-bit	Battery-related value exposed through EC field space
BFCC	16-bit	Full-charge/capacity value used by UPBI()/UPBX()
BADV	16-bit	Battery/adapter voltage used in scaling
MBST	8-bit	Battery status byte used directly by UPBS()
MCUR	16-bit	Current value used by UPBS()
MBRM	16-bit	Remaining energy value used by UPBS()
MBCV	16-bit	Capacity value used by UPBS()
MBTS	1-bit	Selects battery/EC path used by _BST/_BIF/_BIX

What the firmware analysis establishes: the ACPI layer is not an isolated Windows-only abstraction. The decompiled DSDT contains an EC-mediated Smart Battery/SMBus transaction routine (SMRD()) and ACPI battery methods that consume EC-backed fields. Linux captures also recorded an unresolved EC symbol error (AE_NOT_FOUND) and an AML buffer-limit error in WMID methods. The reports explicitly state that causality from those errors to the battery anomaly was not established.

Fault-domain boundary: the established path is BMS/fuel gauge -> SMBus -> EC -> ACPI -> OS. Static DSDT analysis can map this path but cannot uniquely distinguish a failing BMS/fuel gauge from faulty EC-side handling. That is why a known-good battery substitution remains the highest-value discriminator.

10. FIRMWARE-VERSION RECONCILIATION AND REVERSE-ENGINEERING STATUS

The reverse-engineering report is deliberately more conservative than some earlier summaries. It establishes the installed March 20, 2026 F.26 baseline but also notes a newer May 1, 2026 F.26 Rev.A package. The same version-family label does not establish binary identity.

Question	Current status	Implication
Installed firmware baseline	F.26 dated 2026-03-20	Correct experimental baseline to preserve
Newer HP package	F.26 Rev.A dated 2026-05-01	Must be acquired as a separate comparison target
Extracted firmware artifact	7.4 MiB payload acquired	Identity/fingerprint still needs definitive proof
ACPI/EC reverse engineering	Active and credible	Can map interfaces and data paths
Specific causal firmware function	Not established	No final BIOS-root-cause claim should be made
Strongest next firmware result	A/B binary/module/ACPI difference tied to a reproduced fault state	Would substantially strengthen causal attribution

Decision gate: do not overwrite the March baseline before fingerprinting. A controlled A/B comparison of March F.26 against May F.26 Rev.A is a much stronger experiment than general speculation about "BIOS F.26."

This matters for public disclosure because a claim that "F.26 caused the issue" is currently unsupported. The evidence supports a firmware/EC investigation path, not a version-specific causal attribution.

11. MECHANISM SEPARATION AND CAUSAL ASSESSMENT

The evidence should not collapse all battery complaints into one mechanism. Two separate mechanisms are supported by different data streams.

Mechanism	Observed evidence	What it can explain	What it cannot explain alone	Status
A. Telemetry/state mechanism	WMI jumps; Linux near-empty/near-full states; HP diagnostic state flip; Battery changed events	Incorrect SOC/capacity/charging state reporting; low-SOC/AC transition anomalies	Does not identify whether the primary origin is BMS, EC, charging controller, or ACPI handling	Supported hypothesi
B. Power-budget mechanism	Reported hybrid drain under high CPU/GPU demand; adapter ratings; GPU-specific load behavior	Battery discharge while plugged in under heavy load; throttling/power tradeoffs	Does not explain impossible telemetry jumps or OEM diagnostic contradictions	Independe mechanisn supported some case
C. Simple battery wear	85 cycles; near-nominal capacity snapshots; non-monotonic FCC history	Can explain some genuine runtime loss if present	Does not explain capacity increases, abrupt state flips, or same-day diagnostic contradictions	Insufficien sole explanatio
D. Windows-only software issue	Windows reports anomalies	Some OS telemetry presentation	Linux and HP pre-boot diagnostics reproduce state-dependent behavior	Weak as s explanatio

Causal hierarchy

Claim	Assessment	Why
A shared platform-level mechanism deserves investigation	Strongly supported	Cross-GPU recurrence + common EC/ACPI path + cross-platform s behavior
Battery telemetry is unstable on the primary unit	Strongly supported	Raw Windows and Linux state transitions + non-monotonic FCC hi
Exact component is EC firmware	Not established	BMS/fuel gauge and charging path remain unresolved
F.26 caused the anomaly	Not established	Firmware version identity and controlled A/B comparison remain incomplete
All 5600H Victus units are affected	Not established	No controlled prevalence dataset; negative controls exist
Adapter wattage explains all cases	Insufficient	Can explain some hybrid drain but not telemetry discontinuities

12. EXTERNAL CORROBORATION - SEPARATELY LOGGED CASES

Three HP Support Community cases materially strengthen the public context because they show similar battery-state/telemetry phenotypes in the same Victus 15-fb family. They are not treated as proof that all cases share one physical root cause.

Case	Configuration stated by source	Observed behavior	What HP said / what is documented	Analytical
HP Community 9444056	Victus 15-fb0xxx; Ryzen 5 5600H; RTX 3050	12,997 mWh design/full charge reading while actual use remained around 58-59 Wh; battery-change reporting also mentioned	HP Support Agent stated the discrepancy appeared related to EC firmware memory, not battery degradation; recommended full EC reset / BIOS dwell	Very strong corroboratio EC-related t failure mode
HP Community 9539410	Victus 15-fb0xxx; Ryzen 5 5600H; GPU configuration not established in source page	Rapid SOC collapse after unplugging despite normal battery health/diagnostics; persisted after initial support steps	HP support recommended BIOS/chipset/GPU updates, power settings, calibration; user later reported the problem persisted	Independen same-CPU, same-family case
HP Community 9697877	Victus 15-fb0000; exact CPU/GPU not established in source page	Reported FCC about 63.4 Wh -> 15.045 Wh in one day; BIOS Battery Alert 601; UEFI Battery Check Very Weak / failed	Source presents a sudden ~75% reported-capacity change and asks whether this is actual battery damage or misread	Recent sam telemetry discontinuity origin still u

Public-source verification note: the three HP Community cases above were independently checked on 25 August 2026 for this edition. The 9444056 case includes an HP employee response explicitly attributing the 12,997 mWh discrepancy to EC firmware memory. The 9539410 case shows the same broad rapid-drain / rapid-recovery phenotype on a 5600H Victus. The 9697877 case is recent and has exact battery-capacity numbers, but the source page does not establish the CPU/GPU combination, so it is not counted as a 5600H cross-GPU data point.

URLs for the three verified HP Community cases are included in the reference appendix. Additional GPU-specific cases in the cross-GPU matrix remain source-derived from the supplied investigation reports unless separately verified.

13. HIGHEST-VALUE DISCRIMINATING TESTS

Priority	Test	Purpose	Result that changes the causal ranking
1	Known-good compatible battery substitution	Separate battery/BMS/fuel-gauge from motherboard/EC/charging path	Fault disappears -> battery/BMS rises; fault persis EC/charging/motherboard rises
2	Direct SMBus protocol capture	Compare raw fuel-gauge transactions with EC-reported values	Raw gauge correct + EC wrong -> EC handling; ra wrong -> BMS/gauge path
3	Oscilloscope/current-probe rail capture	Separate real hybrid power draw from reporting-only events	Real current event -> physical power path; report -> telemetry/state mechanism
4	Cross-SKU DSDT/BIOS comparison	Identify shared vs configuration-specific ACPI/EC logic	Same method/logic across SKUs -> stronger platf evidence
5	Repeat low-SOC AC transition capture	Test reproducibility under controlled conditions	Repeatable signature raises causal confidence
6	March F.26 vs May F.26 Rev.A binary/ACPI diff	Tie a firmware difference to a fault state	Changed battery/power method + controlled sym correlation -> strongest firmware evidence

The first test is the cleanest practical discriminator. It is also the safest way to avoid prematurely blaming either the battery or the motherboard. The firmware reverse-engineering report independently identifies known-good battery substitution as a high-value discriminator.

14. EVIDENCE INVENTORY AND SOURCE REGISTER

ID	Evidence	Role	Tier
E1	Windows battery-report.html	Historical capacity/SOC timeline; raw HTML	High
E2	battery-report-2026-08-22.html	Fresh F.26 snapshot after anomaly	High
E3	HP UEFI Battery Check screenshots	Manufacturer pre-boot diagnostics	Very High
E4	HP PC Hardware Diagnostics history	PASS -> FAIL progression	Very High
E5	PowerShell WMI BatteryStatus captures	Direct OS-visible battery telemetry	Very High
E6	Fedora UPower/sysfs captures	Independent OS corroboration	Very High
E7	ACAD transition logger	Separates AC detection from charging state	High
E8	DSDT/SSDT decompilation	Firmware architecture mapping	High
E9	Linux kernel ACPI errors	Firmware/ACPI observations	High
E10	Cross-GPU community reports	Pattern confirmation across 5 GPU configs	Medium-High
E11	HP Community 9444056	OEM-confirmed EC memory telemetry case	Very High
E12	BIOS reverse-engineering status	Firmware identity and A/B gate	High
E13	SHA-256 manifests / preserved originals	Integrity preservation	N/A

Primary reference register

[R1] HP Support Community case 9444056 - Victus 15-fb0xxx, Ryzen 5 5600H, RTX 3050; HP Support response identifies the 12,997 mWh discrepancy as an EC firmware-memory issue. https://h30434.www3.hp.com/t5/Gaming-Notebooks/Battery-Capacity-Misread-After-Cleaning-HP-Victus-15-fb0xxx/td-p/9444056

[R2] HP Support Community case 9539410 - Victus 15-fb0xxx, Ryzen 5 5600H; rapid battery drain and recovery after unplugging; user reports persistence after initial troubleshooting. https://h30434.www3.hp.com/t5/Notebook-Hardware-and-Upgrade-Questions/hp-victus-gaming-laptop-15-fb0xxx-battery-drain-issue/td-p/9539410

[R3] HP Support Community case 9697877 - Victus 15-fb0000; reported ~63.4 Wh -> 15.045 Wh capacity transition and HP Battery Alert 601. https://h30434.www3.hp.com/t5/Gaming-Notebooks/HP-Victus-Battery-Suddenly-Dropped-from-63-Wh-to-15-Wh-HP/td-p/9697877

[R4] Supplied HP Victus Forensic Research Report.pdf (18 pages), prepared 24 Aug 2026 - primary source for cross-GPU matrix, DSDT/EC data path, WMI/LINUX/HP diagnostic synthesis, limitations, and forensic test plan.

[R5] Supplied HP Victus Forensic Research Report-1.pdf (6 pages) - condensed cross-GPU comparison and public technical framing.

[R6] Supplied HP Victus Ultimate Consolidated Battery Evidence Report 2026-08-16.pdf - device chronology, raw diagnostic history, Windows battery-report interpretation, Fedora observations, evidence index, and service request.

[R7] Supplied HP Victus BIOS Reverse Engineering Status Report 2026-08-20.pdf - F.26 identity/reconciliation, firmware payload acquisition, and causality gates.

[R8] Supplied ACPI/EC investigation reports and technical evidence reports - DSDT/EC/SMBus architecture and read-only firmware findings.

15. PUBLIC TECHNICAL DISCLOSURE AND QUESTIONS TO HP

This section is written for a public professional audience. It deliberately avoids allegations of intentional misconduct and focuses on reproducible observations, data lineage, and requests for technical answers.

I investigated recurring battery, charging, and power-state anomalies reported on HP Victus systems built around the AMD Ryzen 5 5600H. The analysis compares five GPU configurations - RX 5500M, RX 6500M, GTX 1650, RTX 3050, and RTX 3060 - and combines that external evidence with a read-only firmware/ACPI investigation of a Victus 15-fb0147AX. The strongest device-side result is a repeatable state-transition problem rather than a simple battery-health percentage. Windows WMI captures show remaining-energy reporting changing from 0.300 Wh to about 50.184 Wh around an EC/power reset while charging remained inactive. Fedora independently reproduced near-empty and near-full battery states through a different OS path. HP UEFI diagnostics also changed from normal/pass behavior at higher SOC to Primary Battery: Replace / failed behavior at low SOC. The firmware analysis identifies a concrete BMS -> SMBus -> EC -> ACPI -> OS battery data path. Separately, an HP Support Community case on a Victus 15-fb0xxx with Ryzen 5 5600H and RTX 3050 contains an HP employee response explicitly attributing a 12,997 mWh capacity misread to EC firmware memory. This does not prove that every Victus case has the same cause, but it makes the firmware/EC layer a legitimate engineering target. My current technical conclusion is therefore limited and evidence-led: there is sufficient evidence to justify a platform-level investigation of firmware/EC/ACPI/power-management behavior, but the exact causal boundary between the battery fuel gauge/BMS, EC firmware/state, charging circuitry, and ACPI implementation remains unresolved.

Questions that a technically complete OEM response should address

1. What service diagnosis distinguishes the observed low-SOC failure from simple cell wear?

2. Can HP reproduce the Normal -> Replace diagnostic flip on the same product using a known-good compatible battery?

3. Is the March 20, 2026 F.26 image byte-identical to the May 1, 2026 F.26 Rev.A package for the same product, including ACPI/EC-related modules?

4. Does HP recognize any known issue involving battery fuel-gauge state, EC memory/state, SMBus transactions, or AC-to-battery transitions on this platform family?

5. Can HP explain why AC presence can remain true while Charging=False and ChargeRate=0 at critical SOC in the captured state?

6. If the battery is replaced and the same transition persists, what motherboard/EC/charging diagnosis follows?

Publication discipline: preserve raw screenshots, raw battery-report HTML files, WMI logs, Linux captures, firmware artifacts, timestamps, and SHA-256 manifests alongside this narrative. This report should be updated when the known-good battery substitution or firmware A/B comparison is completed.

End of public technical edition.